

BOAMP Successor Linkage Methodology

Current v3 Benchmark State

BOAMP Data Science Internship Project

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Technical Summary

This report is regenerated from the current v3 benchmark artifacts, not from static notebook output. The latest labelled reference contains 252 labelled anchors and 7,031 anchor-candidate labels. The dev split has 134 anchors and 3,638 pair rows; the validation split has 60 anchors and 1,763 pair rows.

All four linkage algorithms have now been evaluated on the latest v3 reference: `M_A_deterministic`, `M_B_text_ranking`, `M_C_weighted_gated`, and `M_D_fellegi_sunter`. On validation, `M_B_text_ranking` @ 0.70 remains the strongest precision-first method: precision 0.800, recall 0.182, and false-positive rate 0.000. `M_C` has higher recall (0.409) but also more false positives (0.105). `M_D` does not currently challenge the incumbent: precision 0.000 and recall 0.000.

1 Pipeline State

The defensible project pipeline is:

BOAMP notices → standardised notices → procurement episodes
→ candidate pairs → linkage scoring → one successor or abstention → survival dataset.

The candidate-generation rule is unchanged. For an anchor contract with award date A and a candidate with publication date C , the candidate is eligible only when

$$A + 90 \leq C \leq A + 2920.$$

The new v3 benchmark does not replace this candidate pool; it evaluates how well the selection algorithms choose from exposed candidate pairs.

2 Benchmark v3 Reference

The v3 benchmark is national rather than Grand Ouest-only. It samples awarded digital procurement episodes across French region groups and CPV divisions. Only the probability frame supports national weighted estimates; enrichment frames are used for stress cases and diagnostics.

The current annotation caveat remains important: labels are protocol-generated development evidence with double-pass validation and adjudication. They are not official legal renewal ground truth and should not be described as human inter-annotator ground truth.

3 Algorithms

M_A : **deterministic evidence.** Accepts only strong buyer identity, positive CPV continuity, and a minimum text signal. It is conservative but misses many true successors.

M_B : **text ranking.** Ranks same-buyer candidates by TF-IDF cosine similarity and accepts the top candidate only if similarity is at least 0.70. It is transparent and currently best aligned with a precision-first survival event definition.

M_C : **weighted gated score.** Combines buyer, text, CPV, timing, and evidence components, then applies independent gates. It recovers more positives but admits more false links.

M_D : **Fellegi–Sunter probabilistic linkage.** Discretises buyer, text, CPV, and time comparisons, estimates match and non-match level distributions by expectation maximisation, and ranks by posterior match probability. The latest refresh applies the fitted model to the v3 exposure table so this method is no longer skipped.

4 Dev Results

Method	Threshold	Accepted	Precision	Recall	FPR	Coverage
M_A_deterministic	70.000	10	0.600	0.128	0.035	0.075
M_B_text_ranking	70.000	24	0.875	0.447	0.035	0.179
M_C_weighted_gated	70.000	47	0.596	0.596	0.126	0.351
M_D_fellegi_sunter	65.000	4	0.000	0.000	0.023	0.030

Table 1: Unweighted dev metrics on all labelled v3 frames.

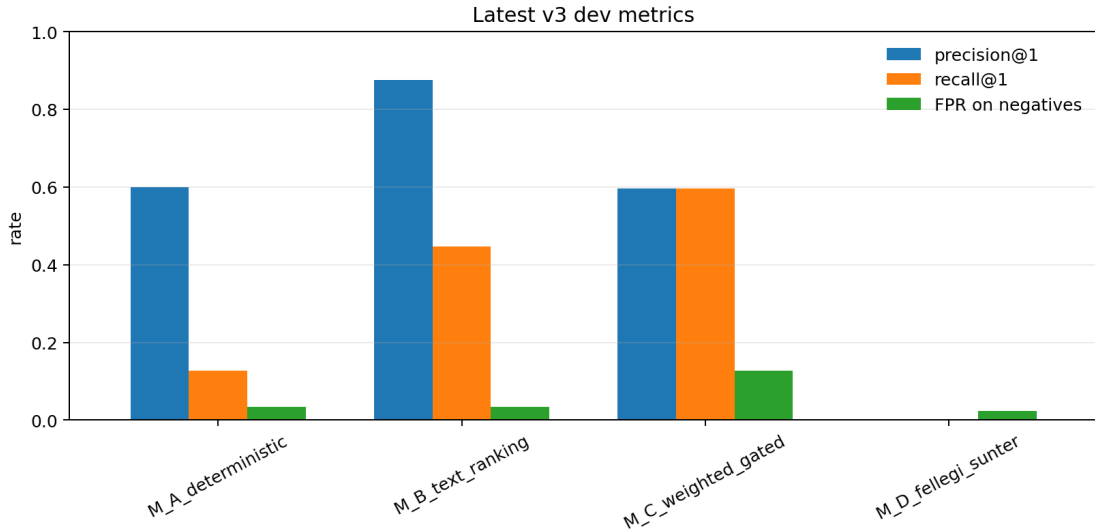


Figure 1: Dev split method comparison generated from the current v3 evaluation JSON.

5 Validation Results

Method	Threshold	Accepted	Precision	Recall	FPR	Coverage
M_A_deterministic	70.000	6	0.167	0.045	0.105	0.100
M_B_text_ranking	70.000	5	0.800	0.182	0.000	0.083
M_C_weighted_gated	70.000	15	0.600	0.409	0.105	0.250
M_D_fellegi_sunter	65.000	1	0.000	0.000	0.026	0.017

Table 2: Unweighted validation metrics on all labelled v3 frames.

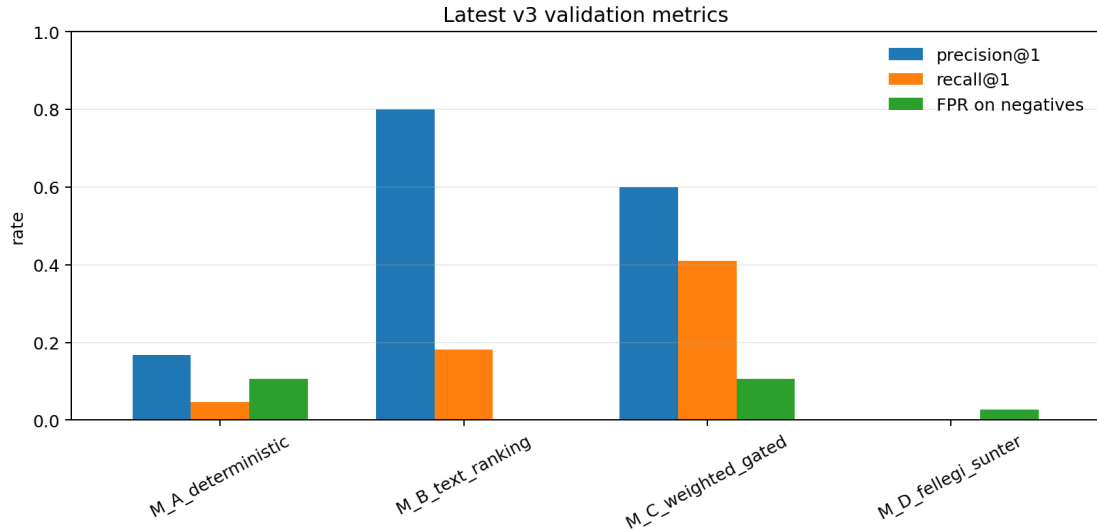


Figure 2: Validation split method comparison generated from the current v3 evaluation JSON.

6 Modeling Tables

The modeling-ready tables include strict, primary, broad, and non-match target columns, plus observable features. They now include the Fellegi–Sunter columns `fs_match_weight` and `fs_match_probability`, so modeling and evaluation use the same feature state.

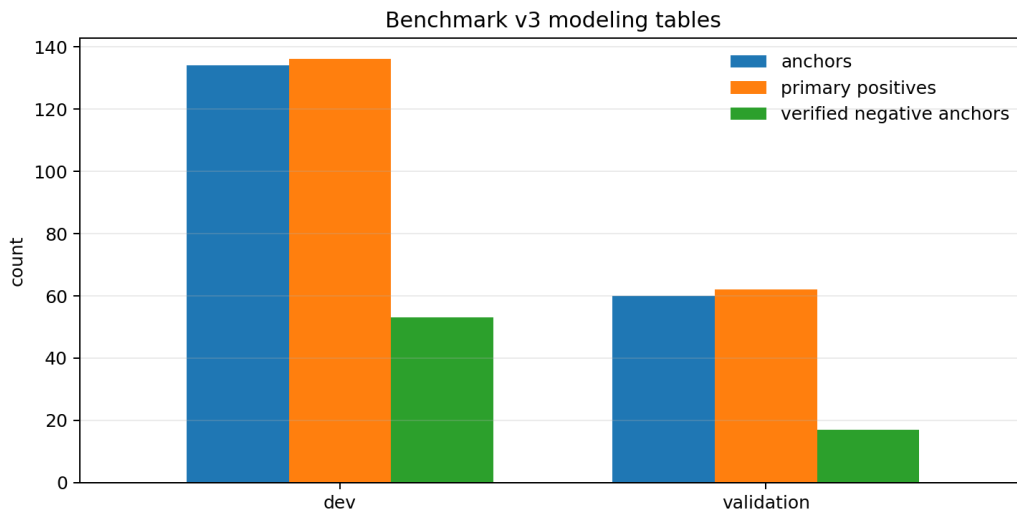


Figure 3: Current v3 modeling table sizes and label support.

7 Decision

The current defensible decision is to keep `M_B_text_ranking @ 0.70` as the primary observable-successor method. It is not the highest-recall method, but the survival analysis is more sensitive to false positives than to abstention: a false positive fabricates both an event and an event time, while an abstention becomes right-censoring.

8 Current Source Files

- `data/processed/boamp_v2/linkage_evaluation_summary_v3_dev_primary.json`
- `data/processed/boamp_v2/linkage_evaluation_summary_v3_validation_primary.json`
- `data/processed/boamp_v2/benchmark_v3/modeling/benchmark_v3_modeling_summary.json`
- `data/processed/boamp_v2/benchmark_v3/benchmark_v3_manifest.json`